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Supermicro X9QRi-F+ Motherboard (Non-SAS Version) – Deep Dive Report

Overview and Introduction

The Supermicro **X9QRi-F+** is a quad-processor server motherboard designed for Intel Xeon E5-4600 series CPUs. It was a flagship platform around 2012–2014, supporting **4 CPU sockets**, extensive memory capacity, and high-bandwidth interconnects for HPC and enterprise applications. There are two main variants of this board – one with an onboard LSI SAS RAID controller (model X9QR7-TF+) and one **without the SAS controller (X9QRi-F+)**, which is our focus. The non-SAS X9QRi-F+ offers the same quad-socket computing capabilities but relies on standard SATA ports (no onboard SAS) and typically uses 1GbE networking instead of 10GbE [1]. This report provides full specifications, compares the SAS vs. non-SAS versions, and dives into known quirks, performance, upgrade tips, optimizations, and user insights for the X9QRi-F+ (non-SAS) motherboard.

Technical Specifications

[2] [3] **Table 1: X9QRi-F+ Key Specifications** (non-SAS variant)

Feature	Description
CPU Support	Quad **LGA2011** sockets supporting Intel **Xeon E5-4600** series (v1 & v2) – up to 12 cores per CPU, 25 MB cache, QPI up to 8.0 GT/s [2] [4]. All four CPUs should be installed to fully enable board functionality (see quirks below).
Chipset	Intel **C602** chipset (PCH) [3]. Provides 6 SATA ports and I/O interfaces. (No SAS controller on this variant's PCH.)
Memory	32 DIMM slots DDR3 (240-pin ECC **Registered** or LRDIMM). Supports up to **1 TB** DDR3 RAM (32×32 GB) [5] [6]. Memory speeds up to 1600 MT/s (and up to 1866 MT/s with v2 CPUs in certain configurations) [7] [8]. Supports 1.5 V or 1.35 V DIMMs; ECC for single-bit error correction [9]. (UDIMMs supported in theory, but RDIMM/LRDIMM recommended for capacity.)
Expansion Slots	**4× PCI Express 3.0** slots: **2× x16** and **2× x8** (physical x16 length for all) [10] [11]. These slots are distributed across the CPUs' PCIe lanes (the board has limited slots due to space consumed by CPUs and DIMMs [12]). Supports add-in cards like GPUs, HBAs, NICs, etc., with the noted lane widths.
Storage (PCH)	**6× SATA ports** from Intel C602: **2× SATA 3.0** (6 Gb/s) + **4× SATA 2.0** (3 Gb/s), supporting Intel RST RAID 0/1/5/10 [13]. No onboard SAS HBA on the X9QRi-F+ (see variant differences). Standard SATA DOM power header provided for boot DOM use [14].

Feature	Description
Networking	**Dual Gigabit Ethernet** (Intel i350 dual-port controller) on the X9QRi-F+ [15]. RJ45 outputs supporting 10/100/1000 Mbps [16]. (By contrast, the SAS-equipped X9QR7-TF+ includes dual 10GBase-T ports – see next section.) Dedicated IPMI LAN port (Realtek RTL8201, 10/100) for management [17].
Onboard Devices	Baseboard Management Controller: Nuvoton **WPCM450** BMC with IPMI 2.0 support (provides IPMI KVM and basic 2D video via an integrated Matrox G200eW graphics core) [18] [19]. Onboard audio: None (server-class board). Super I/O: Nuvoton NCT5573D or W83527HG for legacy I/O [20].
Rear I/O Ports	**Video:** 1× VGA DB-15 port (from BMC) [21]. **LAN:** 2× RJ45 Gigabit LAN + 1× RJ45 IPMI management LAN [22]. **USB:** 4× USB 2.0 ports (rear) [23]. **Serial:** 1× COM1 serial port (rear DB9) [24].
Internal I/O	**USB:** Up to 4 additional USB 2.0 ports via headers – there are front-panel USB headers and one Type-A internal port (total 8 USB 2.0: 4 rear + 3 via front headers + 1 onboard Type-A) [25] [23]. **Serial:** 1× COM2 header. **DOM:** 1× DOM power connector for SATA DOM. **Fan headers:** 10× 4-pin PWM fan headers with monitoring [26]. **TPM header:** present (for optional TPM module).
Form Factor	Proprietary large extended form factor (~16.79″ x 16.4″ / 426 × 417 mm) [27]. This is larger than E-ATX; it requires an appropriate **Supermicro chassis or equivalent**. Common compatible chassis include Supermicro 4U/tower models (e.g. 748 series, 418 series) and some 2U/3U with custom mounting [28]. Ensure the chassis has mounting for this board size and enough clearance for four CPU coolers.
Power	Requires robust PSU with multiple EPS 12V connectors. The board has **four 8-pin EPS12V** power headers (one near each CPU socket) plus the standard 24-pin ATX power [29]. A high-wattage server power supply (with PMBus support for management, if possible) is recommended. Power draw can be significant with 4 CPUs and 32 DIMMs – expect >600 W under full load with high-end CPUs.
BIOS	16 MB SPI Flash EEPROM, AMI BIOS (UEFI 2.3.1 compliant) [30]. Features ACPI 4.0, SMBIOS 2.7, PXE boot, etc. **BIOS version 3.0 (or later)** is required for supporting Ivy Bridge-EP **v2** processors [4]. IPMI firmware is separate (the BMC firmware can be updated to Supermicro's SMT_X9 branch, latest ~v3.20) [31].

Note: The X9QRi-F+ motherboard is essentially the same design as the X9QR7-TF+ (which includes SAS and 10GbE), with certain components not populated. It is a quad-socket board for specialized use; **full functionality (PCIe slots, network, etc.) is only available when all four CPU sockets are populated** [32]. With fewer than 4 CPUs, some PCIe slots and onboard devices may be disabled (per CPU connectivity) or cause firmware hang-ups (see known issues section).

X9QRi-F+ vs. X9QR7-TF+: SAS and NIC Differences

Supermicro produced two main variants of this quad-socket X9 board, often referenced in the same manual: **X9QRi-F+** and **X9QR7-TF+**. The **functional differences** are:

- **Onboard SAS Controller:** The X9QR7-TF+ includes an integrated LSI SAS controller (SAS2208 RAID). This provides **8 SAS2 (6 Gb/s) ports** via two mini-SAS connectors, supporting hardware RAID levels 0,1,5,6,10,50,60 with 1 GB cache [33] [34]. In contrast, the X9QRi-F+ **does not have any SAS controller or SAS ports** on board – it only has the 6 SATA ports from the Intel chipset [35]. (The Supermicro docs note "SAS ports are not included on the X9QRi-F+" [36].) Physically, the SAS model has two SFF-8087 ports and associated jumpers (e.g. JPS1 SAS enable) which are simply unpopulated or absent on the non-SAS board [35].

- **Network Interfaces:** The X9QR7-TF+ comes with **dual 10GbE** NIC ports on board (using an Intel X540 controller, dual 10GBase-T) [37]. The X9QRi-F+ instead features **dual 1GbE** ports (Intel i350 controller) [15]. In other words, the "TF" variant has 10G networking built-in, whereas the "i-F" variant has standard gigabit. (Both have the dedicated IPMI LAN for management.)

- **Product Name Suffix:** The naming reflects these differences – **"7"** in X9QR7-TF+ denotes the SAS (LSI 2208) present, **"T"** indicates 10GBase-T NIC, and **"F"** denotes IPMI support. The X9QRi-F+ ("i" likely indicating standard Intel NIC) lacks the SAS and 10G features [1]. Aside from SAS/NIC, the two boards are otherwise similar in layout, dimensions, CPU/memory support, and expansion slot configuration. They use the same BIOS (with conditional support for the SAS controller on the -7 variant).

- **Onboard Graphics/BMC:** Both variants support IPMI 2.0 via the WPCM450 BMC. Documentation for X9QR7-TF+ sometimes lists an "XGI Volari Z7" video, but in practice the BMC provides a Matrox G200e graphics core on both (the Volari mention may be a data sheet typo – the BMC's G200eW is the active video in either case) [18] [19]. There's no functional difference in terms of graphics or BMC features between the two models; firmware is shared.

Practical impact: If you have the **X9QRi-F+ (no SAS)**, you'll need a discrete HBA/RAID card if you want to attach large numbers of SAS/SATA drives beyond the 6 SATA ports. Conversely, the SAS-equipped X9QR7-TF+ can directly connect up to 8 drives to its onboard RAID (common in Supermicro 4U servers with backplanes). Also, the X9QRi-F+ is limited to 1GbE networking unless you add a 10Gb NIC via PCIe, whereas the X9QR7-TF+ has dual 10G ports ready on board [1]. When planning upgrades or chassis, note that the X9QR7's extra components (SAS connectors, etc.) occupy the board's lower-left corner, whereas the X9QRi-F+ PCB in that area has a cut-out shape – they are mostly cross-compatible in chassis that support one, but the -7 might require slightly more space for its SAS ports. In summary, **X9QR7-TF+ = X9QRi-F+ + (LSI 2208 SAS RAID + dual 10GbE)** [1].

Known Issues, Quirks, and Firmware Considerations

Despite its high-end capabilities, the X9QRi-F+ has some **specific quirks** and best practices to ensure stable operation:

- **All CPUs Populated:** It is **strongly recommended to install four CPUs**. Running this board with only 1–2 CPUs can lead to missing devices or POST hangs. Supermicro's FAQ notes that with only two E5-4600 v2 CPUs, the system would *"regularly hang on POST code 92 or B2"* and that all four CPUs are needed to enable all onboard devices (LAN, etc.) [32]. The architecture ties certain PCIe slots and controllers to specific CPUs; if those CPUs are absent, related functions won't initialize (e.g. PCIe bus lockups at code 92). In short, populate CPU1–CPU4 fully to avoid unpredictable behavior. Partial operation (e.g. two CPUs) might work for basic testing, but expect limited I/O and potential firmware issues in that state.

- **BIOS Version for V2 CPUs:** If upgrading to or using **Ivy Bridge-EP (Xeon E5-4600 v2)** processors, you **must have BIOS version 3.0 or later** installed [4]. Earlier BIOS revisions (2.x) won't recognize v2 CPUs. Since you can't boot an unsupported CPU to flash the BIOS, ensure the board is updated (or have a v1 CPU handy) before installing v2 chips. The latest BIOS at time of writing is 3.0b (released around 2015) which adds v2 support and various fixes. Always reset BIOS settings to default after an update, and reconfigure as needed. The BMC (IPMI) firmware can also be updated (e.g. to SMT_X9_3.20) for improved IPMI web interface and security fixes.

- **Slow Boot and BMC Initialization:** During boot, the system may appear to pause on the Supermicro splash screen for ~20–30 seconds **initializing the BMC/IPMI**. This is normal – the board waits for the management controller to be ready before proceeding (no BIOS prompt will appear until IPMI is up) [38]. If you experience a long hang early in POST with only the logo showing, give it a half minute. (This can be mitigated by disabling "Wait for BMC" in BIOS if such an option exists, but on this class of server it's usually fixed.) Once past that, POST memory testing on 32 DIMMs can also take some time – enabling "Fast Boot" or reducing memory test rigor in BIOS can speed this up at the expense of thorough memory checks.

- **Memory Population Rules:** Follow the manual's guidelines for DIMM installation order. Each CPU has 4 memory channels (8 DIMM slots per CPU, colored in banks). For best results, populate at least **one DIMM per CPU** in the first slot of each channel (the blue slots) [39]. The system can boot with as little as one DIMM total (in CPU1's A1 slot), but that will leave other CPUs without local memory (NUMA imbalance). A recommended minimum is 4 DIMMs (one per CPU) so each CPU has some local memory. Ideally, populate **8 or 16 DIMMs** to engage multiple channels on each CPU (e.g. 2 DIMMs per CPU is a good start, populating two channels on each) [40]. If memory is under-populated (e.g. one CPU missing RAM), the BIOS may disable that CPU or throw memory errors. Also note the **speed downgrade rule**: with **two DIMMs per channel**, DDR3-1866 memory will downclock to 1600 MHz [8]. You only get 1866 MT/s if each channel has a single DIMM; but the increased parallelism of more DIMMs often outweighs the raw MHz loss, so don't be afraid to fully populate for capacity. ECC is always enabled – if you see POST error codes B7/B9 it often indicates a memory issue (reseat modules, check compatibility).

- **PCIe Slot Affinity:** Remember that each PCIe slot is tied to a particular CPU's PCIe lanes. For example, Slot4 might be on CPU1, Slot2 on CPU4, etc. If a CPU is not installed or a CPU is disabled, the slots attached to it won't work. The manual indicates which CPU controls which slot (e.g. "(CPU4) Slot2, (CPU1) Slot4" etc.) – ensure critical add-in cards are placed in slots served by populated CPUs. In a four-CPU config this is moot (all slots active), but in a degraded config keep it in mind.

- **Onboard SAS (for -7 variant) Disabled by Default:** This note is for those with the X9QR7-TF+: the LSI 2208 RAID controller might ship disabled depending on BIOS settings or if fewer CPUs are present. There is a jumper (JPS1) to enable/disable the onboard SAS controller in some revisions [41]. If you have the SAS model and the controller isn't showing at boot, ensure the jumper is set to enable it and all CPUs are present (since it likely hangs off CPU2's PCIe, for example). The non-SAS X9QRi-F+ has no such controller/jumper so this doesn't apply in that case.

- **LSI MegaRAID Firmware Updates (SAS model):** If using the SAS2208 RAID, be aware it's effectively a MegaRAID card on the motherboard. You can update its firmware using LSI tools. Many users choose to **flash the SAS2208 to "IT mode" (SAS2308 HBA firmware)** to use it as an HBA (especially for ZFS or Software-Defined Storage) [42]. This involves cross-flashing to firmware that turns off RAID functionality and makes it a straight SAS controller. Guides for similar X9 boards (e.g. X9DRH-7F) apply here as well [43]. Only attempt this if you're experienced; otherwise use the factory IR (RAID) firmware or simply add a separate HBA if needed for IT mode. (For X9QRi-F+, you'll just add a normal LSI HBA card since there is no onboard SAS.)

- **UEFI, Boot, and NVMe Considerations:** The X9QRi-F+ has an early-generation UEFI BIOS (based on AMI Aptio 4). Notably, **NVMe boot is not natively supported** – when this board was made, NVMe drives didn't yet exist. If you install an NVMe SSD via a PCIe adapter, the BIOS will not see it as a bootable device by default. However, you **can** boot from NVMe with some workarounds: for example, using Clover/DUET boot loaders, or by **modifying the BIOS to insert NVMe driver modules**. Community members have successfully modded similar X9 BIOS to add NVMe support and boot from NVMe SSDs [44]. Alternatively, you can boot from a SATA drive (or USB/DOM) and use NVMe as a data volume. Also, ensure "UEFI Boot" is enabled in BIOS if you plan to boot from GPT disks or newer OSes – the X9's UEFI 2.3.1 firmware should handle Windows Server 2012R2+, Linux, ESXi, etc., but it won't support Secure Boot (predates that feature).

- **"Above 4G Decoding" for GPUs and PCIe Resources:** If you plan to use GPUs (especially with large VRAM) or many PCIe devices, go into BIOS and **enable "Above 4G Decoding"** (sometimes called "PCIe 64-bit Resource Allocation"). This setting allows the BIOS to map devices requiring large memory address spaces (e.g. a GPU with >4 GB VRAM) into the 64-bit address space. Users have reported that a GPU wasn't working until this option was enabled [45]. It's good practice to turn this on in multi-device server builds, as it doesn't harm and will future-proof address allocation.

- **IPMI & Network Quirks:** The IPMI (BMC) shares some features with the host. On this generation, the IPMI can either share one of the LAN ports or have its dedicated LAN – on X9QRi-F+ it has a dedicated Realtek PHY for management [17], so it's separate. If you can't access IPMI: ensure the IPMI LAN cable is in the dedicated port (often labeled or colored differently), or if IPMI is set to failover to LAN1 in settings, use LAN1. The default IPMI login is "ADMIN/ADMIN"; it's advisable to update the BMC firmware and change those credentials for security. Another note: if using the host NICs for heavy traffic, note that on X9 boards the i350 NIC might by default have all offloads on and use part of CPU0's lanes – ensure proper driver config in the OS for performance.

- **Thermals and VRM Throttling:** With four CPUs, each up to 130 W TDP (some models), plus possibly 32 DIMMs, this board will **generate a lot of heat**. It relies on server-style cooling: high airflow through the chassis. All 10 fan headers are PWM controlled; Supermicro's BIOS/IPMI will ramp fans based on CPU temperature. If you run this board on an open bench or in a workstation chassis, be mindful that the voltage regulator modules (VRMs) for the CPUs sit in between sockets and need airflow. There have been cases on multi-socket boards where inadequate airflow causes VRM overheat and CPU throttling. Use proper 4U server heatsinks (for LGA2011 narrow ILM) with high RPM fans or dynatron active coolers if in a tower. In the IPMI web interface, you can set fan mode to "Full" or "Heavy IO" when under load to ensure adequate cooling (auto mode might be conservative). Also consider the power draw: if using a standard ATX PSU, ensure it can supply sufficient 12V amperage on the EPS connectors.

- **Firmware Updates – General:** Check Supermicro's support page for any BIOS or IPMI firmware updates relevant to X9QRi-F+. For example, BIOS 3.0a to 3.0b might fix certain errata or add CPU microcode updates. Similarly, IPMI firmware updates can patch security issues (though WPCM450 BMC is older and not as heavily updated as newer AST2400-based ones). Always follow the Supermicro guidelines for flashing (you can use IPMI flash, DOS/IPMI tool, or EFI shell). After major updates, verify that BIOS settings (especially memory speed, NUMA, power management) are as desired, since they might reset.

Performance and Benchmarking

When it debuted, a fully-loaded X9QRi-F+ was a powerhouse: up to 48 cores / 96 threads (with 4× 12-core Xeon E5-46xx v2) and enormous memory bandwidth from 16 memory channels. It was targeted at **HPC and heavy multi-threaded workloads**. While modern CPUs have surpassed it, it's still useful to understand its performance profile:

- **CPU Compute Performance:** Four E5-4600 v2 CPUs can deliver on the order of **hundreds of gigaflops** of double-precision compute. For instance, a configuration with 4× E5-4620 (8-core 2.2 GHz, total 32 cores) achieves around **130 GFLOPs** in Geekbench's vectorized floating-point tests [46]. Integer multi-core scores in Geekbench 2 were in the ~23k range, and floating-point scores ~45k [47] (for comparison, that outclassed most dual-socket systems of its time). In LINPACK-like workloads, a 4×10-core setup (E5-4650 v2s) can approach roughly ~600 GFLOPs peak theoretical (10 cores * 8 flops per core * 2.9 GHz turbo * 4 CPUs, minus overhead). Real LINPACK results would be lower, but still several hundred GFLOPs – significant for its era. In practice, a single modern CPU (like a 32-core EPYC) can rival this performance, but the quad Xeon can still crunch through parallel tasks decently. Expect excellent scaling on multi-threaded jobs that are NUMA-aware.

- **Memory Bandwidth:** With 4 CPUs, each with four DDR3 channels, the aggregate memory bandwidth is quite high (on paper up to ~170 GB/s if all channels used). Actual measured bandwidth per socket is around 40–45 GB/s. Geekbench "Stream" tests show about **4.3 GB/s per thread** in copy operations [48] [49], which roughly correlates to ~40+ GB/s per CPU. The latency between CPUs (QPI interconnect) is higher than intra-CPU, so NUMA tuning in software is important to maximize throughput. For memory-heavy applications (like large in-memory databases or scientific sims), the board provides both capacity (1 TB) and throughput, though the older DDR3-1600/1866 and QPI links mean it won't compete with newer DDR4/DDR5 platforms in raw bandwidth.

- **I/O and Expansion Performance:** One limitation pointed out by early reviewers is the **scarcity of PCIe slots** relative to the available PCIe lanes [12] [50]. Only four slots for potentially 160 lanes means many lanes go unused – but it's due to board real estate constraints. Each x16 slot can deliver the full 16 GT/s * 16 lanes bandwidth (~32 GB/s) to a device, which is plenty for high-end accelerators or NICs. The x8 slots (~16 GB/s each) are still fine for, say, a 10GbE NIC or a SAS HBA. The **onboard dual 1GbE** (on X9QRi-F+) can collectively handle ~2 Gbps, which is trivial relative to the CPU power; most users will add a faster NIC. If using the SAS model's onboard SAS2208 in RAID mode, note that it's PCIe 3.0 x8 attached – which is ~8 GB/s, far above what spinning disks can do (and enough for a handful of SSDs). **Storage throughput**: The 6 onboard SATA ports (via C602 PCH) share a DMI 2.0 link (approx 2 GB/s total to the chipset). This is not a huge bottleneck unless all 6 ports are saturated by SSDs, in which case ~2 GB/s aggregate is the limit. For better storage performance, an add-in PCIe storage controller or NVMe drives on the CPU's PCIe is ideal.

- **Comparative Notes:** In contemporary terms, a quad E5-4650 v2 (40 cores @ 2.4 GHz) roughly matches a single mid-range modern CPU in multi-core, but falls far behind in single-thread. For example, Technical City's aggregate benchmarks suggest a single Xeon E5-4650 v2 is only ~6–7% of a latest-gen server CPU's performance [51]. So four of them might be ~25% of a top 2023 CPU in throughput. That said, for many homelab or server tasks that can use 40 older cores, the X9QRi-F+ still "feels" very powerful, especially with a balanced config (lots of RAM, maybe some SSDs). It excels at things like parallel compilation, running many VMs/containers, or number-crunching tasks that scale out. It will, however, draw significantly more power doing the same work than a newer platform.

- **Power Consumption:** Speaking of power, fully loaded systems (4 CPUs, high core-count, all RAM slots filled) can idle in the 200–300 W range and push 600–800 W under load. No official power numbers are given by Supermicro, but user measurements in forums for similar 4P E5 systems show ~0.15–0.2 A per CPU at idle (each CPU ~20–30 W idle) and 120–130 W per CPU at max. Plus fans, drives, etc. It's advisable to cap turbo or use power management in BIOS if power or cooling is a concern – or enjoy the winter heating bonus if not . The VRMs are digital and efficient for their time, but still, a 4P board is not built for green computing. Ensure your PSU is 80+ rated and sized appropriately (with some headroom).

- **Benchmarking Tips:** If you run benchmarks, keep an eye on **NUMA affinity** – e.g., something like NUMA-aware linpack or stream can show near-linear scaling across the four sockets, but if a thread on CPU1 is accessing memory on CPU4, performance dips due to QPI latency. For virtualization or OS, enabling NUMA in the hypervisor/OS is important (most modern OSes do). Also, thermals can affect sustained benchmark runs – if you see throttling, check CPU temps and fan speeds. With proper cooling, the CPUs should maintain their turbo frequencies until power limits are hit. The board supports Intel Turbo Boost and EIST; by default these are on, so lightly threaded tasks can boost a single CPU's cores (though "lightly threaded" is relative on a 40-core machine!).

Common Upgrade Paths and Compatibility Tips

One advantage of a board like X9QRi-F+ being a few years old is that **upgrades (CPUs, RAM, etc.) are relatively affordable on the second-hand market**, allowing you to max out its potential. Here are common upgrade and configuration paths, along with compatibility notes:

- **CPU Upgrades:** The board originally may have been used with v1 Xeon E5-4600 (Sandy Bridge-EP, e.g. E5-4650, 8c @ 2.7 GHz). Upgrading to **Xeon E5-4600 v2** (Ivy Bridge-EP, e.g. E5-4650 v2, 10c @ 2.4 GHz or E5-4627 v2, 8c @ 3.3 GHz) can increase core counts and efficiency. Ensure BIOS 3.0+ is installed before installing v2 CPUs [4]. All CPUs in the system must be identical model and stepping for SMP (mixing different models is unsupported). The top-end v2 chips (E5-4667 v2, E5-4657L v2, etc.) can be used if you can source them – for example, 4× E5-4657L v2 (12-core @ 2.4 GHz, 115 W) would give 48 cores/96 threads. Just watch TDP: 4× 130 W CPUs is heavy but within the board's design (it's validated for up to 4× 150 W TDP in some Supermicro chassis with proper cooling). **Note:** The E5-2600 series CPUs are *not* compatible with this board – you must use the E5-4600 series, which have the extra QPI links for 4-socket glueless configuration. Also, E7 Xeons are a different socket (LGA2011-v3 for v2 generation E7s) and won't work here.

- **Memory Scaling:** This platform supports massive memory, and an upgrade here can be transformative for certain use cases. It's capable of **1 TB with 32 GB LRDIMMs (32×32)** [52]. More commonly, you might populate 256 GB (e.g. 16×16 GB) or 512 GB (e.g. 32×16 GB RDIMMs) which is relatively inexpensive now. Make sure to use **DDR3** ECC Registered **or** Load-Reduced **DIMMs (unbuffered consumer DIMMs will not work if more than 2 per channel or in large configs). If you have a mix of sizes or ranks, follow the manual's ordering (typically put highest capacity in slot A1/B1 first). The memory is quad-channel per CPU, so populate evenly across CPUs for balance. Also,** low-voltage (1.35 V) DIMMs** are supported – the board will run them at 1.35 V by default if it detects all LV sticks, which reduces power a bit. If mixing LV and regular, it'll use 1.5 V. For upgrade planning: many started with 8 GB sticks (for ~256 GB total); moving to 16 GB sticks doubles capacity, or going to 32 GB (LRDIMM) doubles again. 32 GB LRDIMMs have dropped in price on resale markets, making 512+ GB builds feasible. Just ensure any LRDIMM you buy is on Supermicro's tested list if possible – not strictly required, but some very high density might need a specific BIOS revision.

- **Storage & HBA Upgrades:** The six onboard SATA ports might not be enough for builds intending to have a lot of drives (e.g. as a storage server). A common upgrade is to add a **SAS HBA** or RAID card in a PCIe slot. For example, an LSI 9211-8i or 9300-8i (IT-mode SAS HBA) can connect to a 24-bay chassis backplane. The PCIe 3.0 x8 slot can easily handle the bandwidth of 8 or more drives. If you have the SAS-equipped X9QR7 board and want to use its onboard controller in IT mode, consider flashing it as mentioned earlier, or simply disable it and use an add-in card if you prefer newer HBA models (the onboard SAS2208 is equivalent to an LSI MegaRAID 9271-8i). For **NVMe SSDs**, since there are no M.2 slots, you'd use PCIe adapter cards. One popular method is a PCIe carrier that holds an NVMe and presents it to the system; these work fine for data drives. Booting from NVMe requires the BIOS mod or a UEFI workaround as discussed. Another upgrade path is adding a SATA/SAS RAID card if a hardware RAID with cache is desired beyond the onboard (for instance, a PCIe NVMe RAID or a higher-gen SAS3108-based card for 12 Gb/s SAS). Ensure any HBA/RAID card has proper airflow (in a 4U chassis usually fine, in a tower consider a slot fan, as those cards can run hot).

- **Network Upgrades:** Out-of-the-box, X9QRi-F+ gives you 2× 1 GbE ports. For many modern use cases, that's a bottleneck. Fortunately, adding **10 GbE (or higher)** is straightforward via PCIe. The board's x8 slots can host, for example, a dual-port 10G NIC (like Intel X520-DA2 for SFP+ or X540-T2 for 10GBase-T). Those typically use PCIe 2.0/3.0 x8 and will operate at full speed (the board's PCIe 3.0 x8 is ample for dual 10G ~20 Gbps). If you need even more, one could install a 40 GbE NIC or Infiniband adapter in an x16 slot. Just keep in mind any large IO cards should be in slots tied to CPU1 or CPU2 ideally (for lowest latency; also CPU1 hosts the IPMI so sometimes dedicating heavy IO on a different CPU can help balance). The built-in IPMI can operate on its dedicated NIC regardless of what you add, or you can configure it to share a port with one of the add-in NICs if desired (but using the dedicated port is simpler). When upgrading NICs, check BIOS for options like "PXE boot" or "iSCSI boot" – you can disable those for any NICs you add if not needed, to save boot time.

- **GPU and Accelerator Upgrades:** Despite being a server board, the X9QRi-F+ can accept GPUs or other accelerators (FPGA cards, etc.) in its PCIe slots. Commonly, people have added GPUs either for GPGPU computing or for running virtual machines with GPU passthrough (e.g. a homelab "all-in-one" server that also runs some gaming VMs or AI tasks). The board physically can fit **two double-width GPUs** (one in each x16 slot) if the chassis has 4U height and enough slots open. For example, in a tower case, you might fit an NVIDIA Tesla or GeForce in the Slot4 (CPU1) and one in Slot2 (CPU4) – note those two slots might be adjacent such that their double-width cards sit next to each other, check the layout. If using large GPUs, ensure a strong PSU (each GPU can draw 200–300W). **Enable Above 4G Decoding** in BIOS to support GPUs with large BARs [45]. Also enable **VT-d (Intel Virtualization for Directed I/O)** in BIOS if you plan on GPU passthrough to VMs. The system does support VT-x/VT-d fully; with 4 CPUs you'll have multiple NUMA nodes for VMs. If you populate GPUs, watch thermals – those cards will dump heat inside the chassis; you may need to run fans at higher speeds. Another consideration: the BMC's VGA will be active by default as the primary video. If you install a GPU and want it to be used by an OS or for compute, no issue – but if you wanted it to be the primary display output (rare on a server), you'd have to set BIOS to prefer the add-in VGA. Most leave IPMI/BMC as the console and use GPUs headless or for passthrough.

- **PSU and Power Delivery:** As mentioned, you'll need a solid power supply setup. In a Supermicro chassis, you'd typically have redundant server PSUs (e.g. 2× 1200W). In a custom setup, a single high-wattage PSU (say 1000–1500W 80+ Platinum) with four EPS12V leads is required. Some workstation PSUs have only two EPS; you might use adapters or dual-PSU configs if trying a non-server PSU (exercise caution doing so). Each CPU's 8-pin must be populated. Also, verify the PSU can handle the combined 12V amperage. If the board is used in a workstation scenario, a UPS is recommended – 4P systems draw a lot at startup (inrush current) and you don't want abrupt outages which could corrupt things.

- **Compatibility Notes:** The X9QRi-F+ is from the Sandy/Ivy Bridge era, which means it uses PCIe 3.0 and DDR3 – generally very compatible with modern OSes (Windows, Linux, VMware ESXi all work fine). Windows 10/11 or Server 2016+ will run, though note Windows 11's installer might complain about "unsupported CPU" – that can be bypassed, as E5-4600 v2 is similar to Ivy Bridge consumer which is just outside Win11's support list. For Linux, virtually any modern distro is fine; just use a kernel new enough to have drivers for the Intel i350 NIC (which is most of them) and for any add-in cards. ESXi 6.x and 7.x support this hardware (the CPUs are VT-x/VT-d capable and on VMware's HCL via OEM systems like Supermicro 8047R). One thing not present is TPM 2.0 (there's a header for an add-on TPM). If you need a TPM for Windows BitLocker or vSphere encryption, you can install a

Supermicro TPM module (e.g. AOM-TPM-9655V for TPM 1.2 or AOM-TPM-9665V for TPM2.0, if supported on X9 – check with Supermicro; X9 might be TPM1.2 only due to BIOS).

Advanced Tuning and Optimization

To get the most out of a quad-socket system, a few tuning steps can be considered:

- **BIOS Settings for Performance:** Enter the BIOS setup (Del key on boot) and review the **CPU and Memory settings**. You'll find options for Intel Turbo Boost, C-States, Power Management, etc. For maximum performance, you can set **Power Performance Mode = Performance** (disables some throttling), enable **Turbo Boost**, and possibly disable CPU C3/C6 states to keep clocks ramped (at the cost of power). Ensure **Memory Mode** is set to **Performance** (NUMA) – usually by default it is NUMA, which is what you want for 4 CPUs (as opposed to Node Interleaving, which would interleave memory across CPUs and generally hurt performance). **Hyper-Threading**: leave this enabled unless you have a specific reason to disable – it generally helps with throughput on these Xeons. The BIOS also may have an option for **High Bandwidth** (enabling above 4G decode, etc.) – as noted, use that for GPUs or large PCIe memory devices. If running Linux, you might also disable the Intel VT-d co-support for interrupts (APICv etc.) if not needed, but usually it's fine to leave on.

- **NUMA and OS Tuning:** In the OS or hypervisor, treat this as a NUMA system with 4 nodes. For example, in VMware ESXi, you'd see 4 NUMA nodes; it's best to size VMs such that they fit in one NUMA node (e.g. a VM with 8 vCPUs and 32 GB RAM will stay within one CPU's resources) to avoid cross-node latency. If you need a big VM, it will span NUMA nodes – ensure the guest OS has NUMA optimization (Windows Server and most Linux with NUMA enabled kernels will do). For HPC applications, use NUMA-aware libraries (MPI, OpenMP) so threads stick to their CPU's memory as much as possible. You can experiment with **CPU affinity** for critical processes to keep them on one socket if they share data heavily. The quad-QPI topology is a **ring** (each CPU connects to two neighbors in a loop), so latency between diagonally opposite CPUs is a bit higher than adjacent ones – some MPI libraries allow topology mapping if that's relevant.

- **Cooling and Fan Optimization:** The BMC controls fans by default using preset curves (Standard, Full, Optimal, HeavyIO). If your system runs hot, switch to **Full** or **Heavy IO** fan mode via IPMI (using ipmitool or the web UI) to ramp all fans to 100%. This ensures stability at the expense of noise. Conversely, if your system is in a lab where noise is an issue and it's running below max load, you can use **Optimal** mode to throttle fans – but keep an eye on temperatures. The CPUs have thermal trip sensors; ideally keep them under ~80°C at full load for longevity. The memory and IOH (PCH) also have sensors – the PCH can get hot if there isn't airflow, so if you see PCH temps > ~85°C, direct a fan at it or improve chassis airflow. The board's many fan headers allow customizing cooling zones (e.g. plug front fans into specific headers to increase airflow near drive bays vs CPU zone). In BIOS, you might find **Fan Speed Control** options (like 4-pin fan speed vs PWM duty cycle); if tuning manually, you can set a static duty cycle. But generally, letting IPMI manage is easiest, just choosing a profile.

- **Stability Tips:** For rock-solid stability, run the system through a burn-in: e.g. memtest86 for memory, a CPU stress test like Prime95 or LINPACK on all 4 CPUs, etc., to ensure cooling and power are sufficient. Quad-socket systems put stress on power delivery; if you have any instability under load (machine check errors, sudden reboots), suspect PSU or VRM overheating. Make sure all power connectors are firmly seated and that no power rail is being overloaded. The board has diagnostic LEDs for each CPU's phase power – check if any show issues (consult manual). In case of BIOS misconfiguration leading to no boot, use the **Clear CMOS jumper (JBT1)** [53] to reset settings. For any weird POST codes, the hex display on the board can be referenced in the manual's code list; common ones: b7/b9 (memory error), 92 (PCI init issue), 19 (IPMI handoff). Keeping firmware up to date also helps stability (newer BIOS microcode can fix errata that cause sporadic issues). The X9QRi-F+ was often used in mission-critical systems (e.g. enterprise servers), so when properly set up, it's very stable for continuous 24/7 use.

- **Use Case Optimizations:** Depending on what you deploy this board for, consider specific tweaks:

- *Virtualization/Hypervisor:* Enable Intel VT-x and VT-d in BIOS (usually default). Use latest VMware tools or Linux KVM with tuning for irqbalance across NUMA, etc. If running many VMs, distribute their CPU affinity to avoid stacking too many on one socket. Also, if using as a NAS + VM combo, isolate storage interrupts to one CPU to improve cache locality.
- *High-Performance Computing:* Leverage Intel's compiler optimizations (ICC/ICX) for Ivy Bridge, and use multithreading libraries that recognize 4 NUMA domains. You might also enable "Cluster on Die" if it were available – but on Ivy Bridge-EP, CoD is not an option (that came with Haswell-EP). So just standard NUMA and maybe disabling logical processors (HT) if the workload doesn't benefit from HT (some HPC codes prefer 1 thread per core).
- *Storage Server:* If using as a storage box (e.g. with TrueNAS or similar), and you added an HBA, consider flashing the BIOS to disable the Option ROM for devices you don't boot from to speed up boot. TrueNAS/ZFS will like the cores and RAM; set ARC size appropriately to use that plentiful memory. Also, if not using the second 1GbE port, you can trunk them or dedicate one for management.
- *Workstation:* If someone uses this as a workstation (rare, but possible), note that Windows will see it as a multi-processor system (grouping cores into processor groups if core count >64, but with 40 or 48 cores it's fine). Some games/apps may not utilize all cores. The board lacks audio, so you'd need a USB audio or sound card. Also, X9 doesn't have USB 3.0, only USB 2.0 – you might add a PCIe USB3 card if needed.

Suitable Use Cases

The Supermicro X9QRi-F+ (and its SAS sibling) are **high-end server boards** well-suited to scenarios that require significant CPU parallelism and memory. Here are some ideal use cases, both originally intended and potential repurposed ones:

- **High-Performance Computing (HPC) & Scientific Simulations:** This board shines in HPC clusters – e.g. computational fluid dynamics (CFD), finite element analysis, Monte Carlo simulations, and other CPU-bound computations. Its quad-CPU design was often used in small cluster nodes or standalone HPC servers. With 32+ cores and large RAM, it can tackle big simulation datasets or number-crunching tasks (weather modeling, seismic processing, etc.). The large memory (up to 1TB) allows in-memory computing for scientific applications [5] [52]. HPC users can also benefit from the option to add specialized NICs (Infiniband or 10/40GbE) for cluster interconnect. Keep in mind HPC codes must be NUMA-aware to fully exploit the 4-socket architecture.

- **Database and In-Memory Analytics Servers:** Quad-socket systems are often deployed for database workloads, especially those needing lots of memory. For example, an in-memory database (Oracle, SAP HANA, etc.) or a large SQL Server instance could utilize 40 cores and hundreds of GB of RAM. The X9QRi-F+ could serve as a hefty database server for an enterprise (though by modern standards, newer platforms offer better per-core performance). It could also be used for big-data processing (Hadoop/Spark nodes) where multiple threads handle data tasks – the high core count and memory help keep such workloads in-memory. The **1TB RAM capacity** is particularly useful for in-memory analytics where dataset fits entirely in RAM.

- **Virtualization Host / Private Cloud:** With many cores and RAM, this board is a good candidate for hosting multiple virtual machines or containers. In a virtualization cluster (using VMware ESXi, Hyper-V, or Proxmox), a single X9QRi-F+ could run dozens of moderate VMs. For example, lab users report hosting VMs for IT labs, game servers, and even as a “central PC for thin clients” for family members [54] [55]. It’s like having a small cloud in one box. vSphere will treat the 4 CPUs as one system (vNUMA to guests as needed). You can allocate say 2–4 cores and 8–16GB RAM to many VMs and still have capacity to spare. The limitation might be I/O if all those VMs need heavy disk or network – so plan add-on NICs and storage accordingly. Additionally, features like VMware FT or large VM support benefit from multiple sockets and memory. Note that licensing (Windows or VMware) can be more expensive with a 4-socket server vs a 2-socket, since many licenses count sockets – for homelab that’s not an issue, but for enterprise, consider it.

- **Cloud Computing Nodes:** Although power-hungry by today’s standards, these boards could be used in OpenStack or cloud clusters for compute nodes. They give a high density of cores per node (especially for earlier OpenStack deployments where per-node core counts mattered). Even Kubernetes/container servers can run a boatload of containers on 48 threads. If used in this manner, one might want to bond the two 1GbE ports or, better, add 10GbE for backend network to avoid network bottlenecks for so many workloads.

- **Server Applications / Enterprise Workloads:** As a general-purpose server, the X9QRi-F+ can handle roles like an application server farm (many app instances on one machine), a heavy web server (though typically web front-ends prefer fewer faster cores – here we have many moderately fast cores), or a multi-purpose on-prem server (running AD, file server, plus some VMs, etc.). It’s an overkill for small tasks but in an enterprise environment it could consolidate many services. Some specific roles: large email server (Exchange with many mailboxes), virtualization of legacy systems, or running multiple Docker services (CI/CD pipelines, etc.) concurrently.

- **High-memory Tasks (RAM Drive, Caching, Simulation):** If you need a **huge memory sandbox**, this board is great. Think of scenarios like: running a 500GB RAM RAM-disk for high-speed caching, or loading entire datasets into memory for crunching. Or running multiple VMs each with tens of GB of memory for testing big software (SharePoint, etc.). The board’s ability to correct single-bit memory errors (ECC) [56] is crucial for stability at high memory densities.

- **Homelab “Monster” Server:** In the enthusiast community, boards like these have been repurposed as homelab servers. For example, homelabbers have built systems to run Plex, game servers, virtualization, and NAS all in one. One Reddit user dubbed it a “monster” and asked what they could do with it [57] – popular answers include running a cluster of VMs for learning, or a combination NAS + hypervisor. Keep in mind the noise and power – a homelab deployment might require some DIY in terms of quiet cooling and acceptance of higher electric bills. But if cores-per-dollar is the metric (purchasing used), these older 4P setups can be very cost-effective for a lab.

- **Not Ideal For:** Conversely, there are some use cases where this board is *not* the best choice: e.g. **single-thread performance intensive tasks** (a modern Ryzen or Xeon would far outperform on per-thread), or **GPU-centric builds beyond 2 GPUs** (limited PCIe slots and no PCIe 4.0/5.0 support). It's also not ideal as an edge server or anything requiring efficiency or low power. And for extremely IO-heavy roles (like a high-end storage array with dozens of NVMe drives), the limited slots could be a bottleneck – a newer platform with more PCIe lanes/slots would serve that better.

In summary, the X9QRi-F+ is suitable for roles that need **lots of CPU threads and memory** in a single system – HPC, VM consolidation, big-memory apps – and less suited for roles that need extreme per-thread speed or massive expansion. Even today in 2025, a well-configured X9QRi-F+ can handle many server workloads admirably, especially in labs or smaller scale deployments, thanks to its robust build and enterprise features.

Community Insights and User Experiences

Insights from **community forums, reviews, and Q&A** can be invaluable for understanding real-world behavior of this board. Here are some noteworthy points gathered from user discussions on platforms like Reddit, ServeTheHome, and TrueNAS communities:

- **POST and Debugging:** Users who experienced POST issues learned the hard way that all CPUs and a proper memory configuration are needed. One Reddit homelabber was stuck at a POST code and later found out they needed either all four CPUs or had to adjust the memory arrangement. Supermicro's official guidance confirms populating all sockets to enable onboard devices [32]. Another user shared that the board *"hangs on B7 [memory initialization]"* if the DIMM population isn't per guidelines – once corrected, it booted fine. The takeaway: if you get mysterious POST codes (B7, B9, 92, etc.), double-check CPUs and memory seating/order first.

- **Memory Configuration Questions:** On ServeTheHome forums, a prospective buyer asked how few DIMMs they could use to start with a X9QRi-F+. The community recommended at least one DIMM per CPU, but ideally two per CPU in the first banks for balance [40]. They also clarified that **populating all blue slots (i.e. 1A, 1B on each CPU)** is a good strategy and that using multiple DIMMs per channel will cap speed at 1600 MHz [8]. One member noted that DDR3 RDIMMs have become very cheap, encouraging loading up the RAM to take advantage of the system [58]. This reflects a general consensus: maximize the RAM if you can – the platform was designed to handle it, and many use-cases (VMs, caches) will gladly use the memory.

- **"Above 4G Decoding" fix:** A Reddit user who installed a GPU found the system wouldn't properly initialize the GPU until they enabled Above 4G Decoding in the BIOS [45]. This is a common tweak on older boards when using modern GPUs with large BAR sizes or when populating large PCIe memory spaces (like multiple GPUs or NVMe adapters). Community tip: always flip that setting to "Enabled" when building a system like this with any 64-bit PCIe devices – it prevents a lot of headaches.

- **Noise and Thermal Management:** Some homelab posts mention that the stock Supermicro cooling solutions can be *very loud* in a home environment. One user building a VM server with X9QRi-F+ planned to put it in a rack in a separate room due to noise [54] [59]. They sought advice on quieting it – suggestions included using aftermarket cooling if possible or running the board in optimal fan mode. However, most community members acknowledge that a 4P board will never be "PC quiet" under load; a closet or basement rack is the way to go. If attempting a quieter setup, large slow fans and possibly custom ducting might be needed to ensure all CPUs get airflow even at lower RPMs. The BMC's fan control can sometimes be tweaked via IPMI tool to custom duty cycles as well.

- **IPMI Usage:** Many forum users praise the convenience of IPMI on this board – you can do full remote KVM, power control, and even BIOS updates via the IPMI web interface. Some did note that the older WPCM450 BMC is not as fast or fancy as newer ASpeed chips (the web interface is quite old-school, and KVM redirection can be clunky on modern browsers). But it gets the job done for headless management. A TrueNAS forum post reminded that **to configure IPMI network settings, you do it either via BIOS or the IPMI default (which is DHCP)** – the user was initially confused trying to set an IP in BIOS where there was no option, forgetting the IPMI has its own config utility or defaults [60]. So community tip: use the **ipmitool** utility or the default credentials to connect and set a static IP if needed.

- **Firmware and Support:** As this board is several generations old, official updates have slowed. Community members have stepped in to provide solutions, like BIOS modding for NVMe or sharing tips on cross-flashing the LSI RAID. On forums like ServeTheHome, enthusiasts shared modded BIOS binaries (though caution is advised using those – better to DIY or trust sources). One user successfully modded the BIOS to get a 1TB Samsung 960 Pro NVMe bootable in a X9DR (dual-socket cousin) [61] and others followed suit, confirming it works on X9 boards [44]. This collective knowledge essentially extends the useful life of the platform by adding modern capabilities.

- **Longevity and Reliability:** There are reports of these boards running for many years continuously. Supermicro's build quality is generally high – solid capacitors, good VRMs. A few users did mention that after ~5-7 years, they needed to replace the CR2032 CMOS battery (typical maintenance). One should also check for any **bulging capacitors** or dust in the heatsinks if buying a decommissioned unit – standard refurb practices. The community generally regards X9 series as very reliable. In fact, some prefer X9 quad-socket over newer X10 quad-socket (which were E7-based and far more expensive) because the X9 with E5-4600 v2 hit a sweet spot of performance per dollar on the used market.

- **Power Consumption in Labs:** Home users who measured power draw often share numbers so others know what to expect. For instance, a user with 4× E5-4650 (8-core, 130W each) noted an idle power around 250W and full load nearly 700W with drives and fans. Some mitigations from community: enable CPU C-states in BIOS to drop power at idle (if you don't mind latency); undervolt or use power-efficient CPU models (like the "L" suffix Xeons) – e.g. E5-4650L v2 (10-core 2.5 GHz 115W) instead of E5-4650 v2 (10-core 2.4 GHz 130W) to shave some watts. Also, spinning down idle drives, and using fan control profiles can help a bit. These boards also allow disabling of some controllers (e.g. if you are not using USB 2.0 ports or the onboard video after boot, disabling those might save marginal power).

- **Real-world tasks and satisfaction:** Many homelab posts essentially boil down to *"I got this board (or server) cheap – what cool stuff can I do with it?"*. After building out, users report being able to run large lab environments. For example, running an entire AD/DNS/DHCP/Exchange stack in VMs, alongside a SAN/NAS VM, and maybe a CI/CD pipeline – all on one machine – is feasible and has been done. The main complaint is usually heat/noise, not capability. One user humorously said after assembling a quad-Xeon system that it "doubles as a space heater and hair dryer," so they only run it in winter to warm the house while folding proteins. So community consensus: **the X9QRi-F+ is a beastly board that, when used to its strengths, remains quite useful**, but it demands proper environment (power, cooling) and is impractical for light or quiet workloads.

In conclusion, the Supermicro X9QRi-F+ (non-SAS) is a highly capable quad-socket board that can still deliver strong performance in multi-threaded and memory-intensive applications. Official documentation and reviews highlight its robust specs (quad CPUs, 32 DIMMs, etc.) while community experiences provide guidance on how to run it optimally (populate all CPUs/RAM, update the BIOS, tweak BIOS for new hardware, and be mindful of power/cooling). Though replaced in the market by newer generations, it remains a relevant platform for specific needs – a testament to Supermicro's engineering and the lasting utility of a well-balanced multi-processor system.

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